From First Principles to the LBM Solver

Maxwell's equations → vector-potential magnetostatics → the f, g, h lattices

MTP — Negative rotational viscosity in ferrofluids via phase-field MRT-LBM

The map of this document

There are three "worlds" the simulation couples, each with its own lattice:

- **g-lattice** — *where is the fluid?* Tracks the phase field ϕ (droplet vs. carrier).
- **f-lattice** — *how does it move?* Tracks momentum: Navier–Stokes with forces.
- **h-lattice** — *what is the magnetic field?* Solves magnetostatics for A_z.

Threaded through them is the **magnetization law** (MRSh), which says how the particles' magnetic moments respond to field and to being spun by the flow. We build each from its governing PDE down to the exact equilibrium coded in the solver.

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1 The magnetic field: Maxwell → vector potential

1.1 Quasi-static Maxwell

On the timescales of the flow the field is quasi-static. Ampère's law without displacement current and the absence of magnetic monopoles give, at every instant,

$$\nabla \times \mathbf{H} = 0, \qquad \nabla \cdot \mathbf{B} = 0, \qquad \mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}). \tag{1}$$

1.2 The 2D vector potential

In two dimensions with fields in the plane, introduce a single scalar $A_z(x, y)$ via

$$\mathbf{B} = \nabla \times (A_z \hat{z}) = \big(\partial_y A_z,\ -\partial_x A_z\big). \tag{2}$$

Then $\nabla \cdot \mathbf{B} \equiv 0$ automatically. Take the curl of $\mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}$ and set it to zero:

$$0 = (\nabla \times \mathbf{H})_z = \frac{1}{\mu_0}(\nabla \times \mathbf{B})_z - (\nabla \times \mathbf{M})_z = -\frac{1}{\mu_0}\nabla^2 A_z - (\nabla \times \mathbf{M})_z, \tag{3}$$

giving the governing magnetostatic equation and the field recovery:

$$\boxed{\nabla^2 A_z = -\mu_0 (\nabla \times \mathbf{M})_z}, \qquad \mathbf{H} = \frac{1}{\mu_0}\big(\partial_y A_z,\, -\partial_x A_z\big) - \mathbf{M}. \tag{4}$$

Why a curl source, not a divergence

The scalar-potential version had $\nabla^2\psi = \nabla\cdot\mathbf{M}$ — the field is sourced by where magnetization *piles up* (divergence). The vector-potential version has a *curl* source: the field is sourced by where magnetization *circulates*. Same Poisson operator, same lattice, same relaxation — only the right-hand side and the recovery formula change. That is the entire cost of the pivot (see PDF 1).

1.3 Scalar vs. vector potential — what actually changed

	Scalar ψ (old)	**Vector A_z (current)**
Field ansatz	$\mathbf{H} = -\nabla\psi$	$\mathbf{B} = \nabla \times (A_z \hat{z})$
Poisson equation	$\nabla^2\psi = \nabla \cdot \mathbf{M}$	$\nabla^2 A_z = -\mu_0(\nabla \times \mathbf{M})_z$
$\nabla \cdot \mathbf{B} = 0$	imposed	automatic
Field recovery	$\mathbf{H} = -\nabla\psi$	$\mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}$
Mean field along x	**lost** (topology)	carried by wall BC
Equilibrium	$h_k^{eq} = w_k\psi$	$h_k^{eq} = w_k A_z$

2 The magnetization law (MRSh effective field)

The particle moments do not follow the field instantly — they relax with Brownian time τ_B, and the flow's vorticity tips them. The Martsenyuk–Raikher–Shliomis (MRSh) model, in the effective-field variable $\boldsymbol{\xi}_e$, is (Bacri et al., Eqs. 5–6):

$$\mathbf{M} = \phi\, M_s\, L(\xi_e)\, \hat{\mathbf{e}}, \qquad \hat{\mathbf{e}} = \boldsymbol{\xi}_e/\xi_e, \qquad L(x) = \coth x - \tfrac{1}{x}, \tag{5}$$

$$\frac{d\boldsymbol{\xi}_e}{dt} = \boldsymbol{\Omega} \times \boldsymbol{\xi}_e - \frac{1}{\tau_B}\frac{L(\xi_e)}{L'(\xi_e)}\Big(1 - \frac{\xi_\parallel}{\xi_e}\Big)\hat{\mathbf{e}} + \frac{1}{2\tau_B}\Big(\frac{\xi_e}{L(\xi_e)} - 1\Big)\boldsymbol{\xi}_\perp, \tag{6}$$

where $\boldsymbol{\xi} = \frac{\xi_0}{H_0}\mathbf{H}$ is the equilibrium Langevin argument, $\xi_\parallel = \hat{\mathbf{e}}\cdot\boldsymbol{\xi}$, $\boldsymbol{\xi}_\perp = \boldsymbol{\xi} - \xi_\parallel\hat{\mathbf{e}}$, and $\boldsymbol{\Omega} = \frac{1}{2}(\nabla\times\mathbf{u})_z\hat{z}$ is half the vorticity.

What each term does

$\boldsymbol{\Omega} \times \boldsymbol{\xi}_e$: the **flow spins the moment**. The $\hat{\mathbf{e}}$ term: **longitudinal relaxation** — the magnitude relaxes toward the Langevin value with time $\tau_\parallel = \tau_B\, \xi_e L'/L$. The $\boldsymbol{\xi}_\perp$ term: **transverse relaxation** — the direction swings toward the field with time $\tau_\perp = 2\tau_B L/(\xi_e - L)$. Negative viscosity is born from the *competition* between spinning and relaxing: if the field oscillates at the right rate, the lag makes the magnetic torque *help* the flow instead of resisting it.

Because $H_z \equiv 0$ and vorticity is purely out-of-plane, $\xi_{e,z} = 0$ is an *exact invariant* of Eq. (6) — not an approximation. The planar channel is a genuine realization of Bacri's 3D theory, and the 3D Langevin function is retained. The rotation term reduces to $(\boldsymbol{\Omega} \times \boldsymbol{\xi}_e) = (-\Omega\, \xi_{e,y},\ \Omega\, \xi_{e,x})$.

Saturation magnetization is fixed by the physical scaling $\xi = \mu_0 m H / k_B T$ and $\tau_B = 3\eta V / k_B T$:

$$M_s = \frac{3\eta_h \phi_p\, \xi_0}{\mu_0 \tau_B H_0} = \phi_p M_d. \tag{7}$$

Numerically the ODE is integrated with RK4 (8 sub-steps per Δt), $\mathbf{H}$ interpolated linearly between steps and Ω frozen; small-ξ series expansions of L, L' avoid catastrophic cancellation below $|\xi| < 0.2$.

3 The magnetic body force

The field acts on the fluid through two pieces:

$$\mathbf{F}_m = \underbrace{\mu_0 (\mathbf{M} \cdot \nabla)\mathbf{H}}_{\text{Kelvin}} + \underbrace{\tfrac{1}{2} \nabla \times \big[\mu_0 (\mathbf{M} \times \mathbf{H})\big]}_{\text{antisymmetric stress}}. \tag{8}$$

The Kelvin force pushes the fluid toward stronger field (it is what pumps the flow in PDF 4). The antisymmetric-stress term is the torque density from the misalignment of $\mathbf{M}$ and $\mathbf{H}$ — the seat of rotational viscosity. In 2D, with $T = \mu_0 (M_x H_y - M_y H_x)$,

$$\mathbf{F}_m^{(2)} = \tfrac{1}{2} \big(\partial_y T,\ -\partial_x T\big). \tag{9}$$

4 The phase field: conservative Allen–Cahn (g-lattice)

The interface is tracked by $\phi \in [0, 1]$ obeying the conservative Allen–Cahn equation:

$$\partial_t \phi + \nabla \cdot (\phi \mathbf{u}) = \nabla \cdot \Big[M_\phi \Big(\nabla \phi - \tfrac{4}{D} \phi (1 - \phi) \hat{\mathbf{n}} \Big) \Big], \qquad \hat{\mathbf{n}} = \frac{\nabla \phi}{|\nabla \phi|}. \tag{10}$$

The D2Q9 MRT equilibrium that produces this is

$$\boxed{g_k^{eq} = w_k \phi \Big[1 + \frac{\mathbf{c}_k \cdot \mathbf{u}}{c_s^2} \Big] + w_k \frac{M_\phi}{c_s^2} \frac{4}{D} \phi (1 - \phi)\, \mathbf{c}_k \cdot \hat{\mathbf{n}}}, \qquad \phi = \sum_k g_k, \tag{11}$$

with collision $g_k^* = g_k - [\mathbf{M}^{-1} \mathbf{S}_g \mathbf{M} (g - g^{eq})]_k$ and $\tau_\phi = M_\phi / (c_s^2 \Delta t) + \tfrac{1}{2}$. The chemical potential, surface-tension force and density follow as

$$\mu_\phi = 4\beta \phi (\phi - 1)(\phi - \tfrac{1}{2}) - \kappa \nabla^2 \phi, \quad \mathbf{F}_s = \mu_\phi \nabla \phi, \quad \rho = \rho_l + \phi (\rho_h - \rho_l), \tag{12}$$

with $\beta = 12\sigma / D$, $\kappa = 3 D \sigma / 2$.

5 The hydrodynamics: pressure-based MRT (f-lattice)

The target is incompressible Navier–Stokes with all body forces:

$$\nabla \cdot \mathbf{u} \approx 0, \qquad \rho (\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla p + \nabla \cdot [\eta (\nabla \mathbf{u} + \nabla \mathbf{u}^\top)] + \mathbf{F}_s + \mathbf{F}_m + \mathbf{F}_b. \tag{13}$$

Using the incompressible equilibrium (with Γ_k the standard second-order Maxwellian weight),

$$\Gamma_k = w_k \Big[1 + \frac{\mathbf{c}_k \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{c}_k \cdot \mathbf{u})^2}{2 c_s^4} - \frac{u^2}{2 c_s^2} \Big], \qquad \boxed{f_k^{eq} = w_k p + \rho c_s^2 (\Gamma_k - w_k)}, \tag{14}$$

the forcing term (Guo-type, adapted to the pressure formulation) is

$$F_k = (\mathbf{c}_k - \mathbf{u}) \cdot \left[c_s^2 \nabla\rho\,(\Gamma_k - w_k) + \Gamma_k\,\mathbf{F}_{\text{tot}}\right], \qquad \mathbf{F}_{\text{tot}} = \mathbf{F}_s + \mathbf{F}_m + \mathbf{F}_b. \tag{15}$$

Collision and macroscopic recovery:

$$f_k^* = f_k - [\mathbf{M}^{-1}\mathbf{S}_f\mathbf{M}(f - f^{eq})]_k + [\mathbf{M}^{-1}(\mathbf{I} - \tfrac{1}{2}\mathbf{S}_f)\mathbf{M}F]_k, \tag{16}$$

$$\mathbf{u} = \frac{\sum_k \mathbf{c}_k f_k}{\rho c_s^2} + \frac{\Delta t}{2\rho}\mathbf{F}_{\text{tot}}, \qquad p = \sum_k f_k + \frac{\Delta t}{2} c_s^2\,\mathbf{u}\cdot\nabla\rho. \tag{17}$$

The magic-parameter wall trick

Viscosity is set by s_ν; the "ghost" rates are tied to it by $s_q = 8(2 - s_\nu)/(8 - s_\nu)$. This fixes $\Lambda = (\frac{1}{s_\nu} - \frac{1}{2})(\frac{1}{s_q} - \frac{1}{2}) = \frac{3}{16}$, which places the bounce-back wall *exactly* halfway between nodes — so a Poiseuille parabola lands on the analytic profile with no wall-position error. This is why static agreement reaches 0.3%.

The relaxation rate is interpolated harmonically in viscosity across the interface, $\frac{1}{\tau} = \frac{1-\phi}{\tau_l} + \frac{\phi}{\tau_h}$, $\tau_{l,h} = \eta_{l,h}/(\rho_{l,h} c_s^2 \Delta t)$, $s_\nu = 1/(\tau + \frac{1}{2})$.

6 The magnetostatics: A_z in pseudo-time (h-lattice)

Equation (4) is solved by driving a diffusion in pseudo-time s to steady state:

$$\partial_s A_z = \varepsilon\nabla^2 A_z + \varepsilon\left[\mu_0(\nabla\times\mathbf{M})_z + r\right], \tag{18}$$

run for N_{sub} sub-steps per physical step with $\mathbf{M}$ frozen. The D2Q9 form uses the simplest scalar equilibrium and recovers the field from the *first moment* via Chapman–Enskog:

$$\boxed{h_k^{eq} = w_k A_z}, \qquad A_z = \sum_k h_k, \qquad \tau_h = \frac{\varepsilon}{c_s^2\Delta t} + \frac{1}{2}, \tag{19}$$

$$B_x = -\frac{\sum_k c_{ky} h_k}{\tau_h c_s^2 \Delta t}, \qquad B_y = \frac{\sum_k c_{kx} h_k}{\tau_h c_s^2 \Delta t}, \qquad \mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}. \tag{20}$$

The wall carries the Neumann data $\partial A_z/\partial n = \gamma = \mu_0(\mathbf{H}_{\text{app}} + \mathbf{M}_w)\cdot\hat{\mathbf{t}}$, injected as

$$h_k(\text{wall}) = h_{\bar{k}}^* + 6\,\varepsilon\,\gamma\,w_k\Delta t, \qquad \bar{k} = \text{specular in } y,$$

the factor $6 = 1/\sum_{c_{ky}>0} w_k$ matching injected flux to diffusive flux. The Neumann problem is solvable only if the net source vanishes; the compatibility constant r enforces $\oint$ balance and telescopes to $r \approx 0$ to round-off. $\int A_z$ is conserved — no drift.

Why finite differences beat the first moment for H

Recovering $\mathbf{H}$ from $\sum_k \mathbf{c}_k h_k$ assumes near-equilibrium populations. For a field with $\partial_x\partial_y A_z \neq 0$ (which the pumping field has), that assumption gives H_y errors of order k. Recovering $\mathbf{H}$ instead from *finite differences of* A_z is exact — in the $\mathbf{M} = 0$ test the lattice reproduced A_{app} to 10^{-15}.

7 How it all fits together — one timestep

1. $\phi = \sum g$; compute $\hat{\mathbf{n}}$ and μ_ϕ.
2. Predictor $\mathbf{u}^*$ from previous force; $\Omega = \frac{1}{2}\nabla \times \mathbf{u}^*$.
3. Recover $\mathbf{H}$ from h and $\mathbf{M}^{n-1}$; RK4 integrate MRSh $\rightarrow \mathbf{M}^n$.
4. Source S_A and r; run N_{sub} h-sub-steps with the current applied field.
5. Store `H_prev`.
6. Assemble $\mathbf{F}_m(\mathbf{M}^n, \mathbf{H})$, $\mathbf{F}_s$, ρ.
7. Update $\mathbf{u}$ and p.
8. Build g^{eq}, f^{eq}, F_k; collide g, f; stream all lattices.

$g \rightarrow \phi \;\Rightarrow\; \rho, \eta$; $\quad f \rightarrow \mathbf{u} \;\Rightarrow\; \Omega$; $\quad \Omega + \mathbf{H} \;\Rightarrow\; \mathbf{M}$ (MRSh); $\quad \mathbf{M} \;\Rightarrow\; h \rightarrow \mathbf{H}$; $\quad \mathbf{M}, \mathbf{H} \;\Rightarrow\; \mathbf{F}_m \;\Rightarrow\; f$. The loop closes: flow tilts the magnetization, magnetization sets the force, the force drives the flow.